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class:SY. BSc. IT *italicized text*

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

```
import pandas as pd

data = {
  "name": ["Raj", "Kaushal", "Amit"],
  "age": [92, 42, 88],
  "qualified": [True, False, False]
}
df = pd.DataFrame(data)

newdf = df.filter(items=["name", "age"])
print(newdf)
```

```
      name  age
0      Raj   92
1  Kaushal   42
2     Amit   88
```

5b. Write a python code to show statistical information on given data set.**bold text bold text**

```
import statistics

print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))
print(statistics.mean([1, 3, 5, 7, 9, 11]))
print(statistics.mean([-11, 5.5, -3.4, 7.1, -9, 22]))
print("raj")
```

```
7
6
1.8666666666666667
raj
```

statistics.median()

```
import statistics

print(statistics.median([17, 9, 3, 8, 2, 1, 11]))
print(statistics.median([1, 3, 5, 7, 9, 7]))
print(statistics.median([-11, 5.5, -3.4, 7.1, -9, 22]))
```

```
8
6.0
1.05
```

statistics.mode()

```
import statistics

print(statistics.mode([1, 5, 3, 4, 7, 3, 9, 11]))
print(statistics.mode([1, 1, 9, -5, 7, -9, 11]))
print(statistics.mode(['Raj', 'Sunny', 'Amit', 'rani']))
```

```
3
1
Raj
```

statistics.stdev()

```
import statistics

print(statistics.stdev([1, 3, 5, 6, 4, 15]))
print(statistics.stdev([2, 2.5, 1.75, 3.1, 1.75, 2.8]))
print(statistics.stdev([-11, 5.5, -3.1, 7.1]))
print(statistics.stdev([1, 30, 50, 200]))
```

```
4.88535225614967
0.5697952848757759
8.380682152028756
88.80831417534434
```

statistics.variance()

```
import statistics

print(statistics.variance([1, 2, 6, 7, 9, 11]))
print(statistics.variance([2, 2.5, 1.4, 3.1, 1.75, 2.8]))
print(statistics.variance([-11, 5.5, -3.4, 7.1]))
print(statistics.variance([1, 30, 50, 50]))
```

```
15.2
0.42441666666666666
70.80333333333333
536.9166666666666
```

5c.Write a python code to create pandas series from dictionaries.

```
import pandas as pd
dictionary = {'Raj': 45, 'Amit': 25, 'nitesh': 39}

# create a series
series = pd.Series(dictionary)

print(series)
print("Raj")
```

```
Raj       45
Amit      25
nitesh    39
dtype: int64
Raj
```

```
import pandas as pd

dictionary = {'D': 23, 'B': 55, 'C': 30}

# create a series
series = pd.Series(dictionary)

print(series)
print("Raj")
```

```
D    23
B    55
C    30
dtype: int64
Raj
```

```
import pandas as pd
dictionary = {'Raj': 50, 'Baby': 10, 'Kaushal': 80}

series = pd.Series(dictionary, index=['Baby', 'Chandani', 'Amit',])

print(series)
```

```
Baby        10.0
Chandani     NaN
Amit         NaN
dtype: float64
```

```
import pandas as pd

dictionary = {'A': 32, 'B': 88, 'C': 80}

series = pd.Series(dictionary, index=['B', 'C', 'D', 'A'])

print(series)
print("Raj")
```

```
B    88.0
C    80.0
D     NaN
A    32.0
dtype: float64
Raj
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays

```
import pandas as pd
dict = {'name':["RAj", "Amit", "sudhir", "Geeku"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score':[90, 40, 80, 98]}

df = pd.DataFrame(dict, index = [True, False, True, False])
print(df.loc[True])
```

```
        name  degree  score
True     RAj     MBA     90
True  sudhir  M.Tech     80
```

```
import pandas as pd

dict = {'name':["Raj", "Kaushal", "Amit", "Geeku"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score':[11, 40, 40, 96]}

df = pd.DataFrame(dict, index = [True, False, False, False])


print(df.loc[False,])
```

```
          name  degree  score
False  Kaushal     BCA     40
False     Amit  M.Tech     40
False    Geeku     MBA     96
```